

Standard form

Worksheet 2 (Calculator Based)

1. S10/qp22/q3

Planet	Mass (kg)	Diameter (km)
Mercury	3.30×10^{23}	4880
Venus	4.87×10^{24}	12 100
Earth	5.97×10^{24}	12 800
Mars	6.42×10^{23}	6790

The mass and diameter of the planets in the inner solar system are shown in the table.

- List the planets in order of mass, starting with the lowest. [1]
- Find the radius, in kilometres, of Mars, giving your answer correct to 1 significant figure. [1]
- Giving your answer in standard form, find the total mass, in kilograms, of Venus and Mars. [1]
- [Volume of a sphere = $\frac{4}{3}\pi r^3$]. Giving your answer in standard form, find the volume, in cubic kilometres, of the Earth. [2]

2. **S17/qp22/q2**

The table below shows the population, given to the nearest thousand, of some countries.

Country	Population in 2014	Population in 2015
Pakistan	185 133 000	188 169 000
China	1 393 784 000	1 402 007 000
South Korea	49 512 000	49 765 000
Thailand	67 223 000	67 438 000

- a. In 2015, how much larger was the population of Pakistan than the population of South Korea? [1]
- b. Which country had the smallest increase in population between 2014 and 2015? [1]
- c. Write the population of South Korea in 2014 in standard form. [1]
- d. Find the percentage increase in population of Pakistan from 2014 to 2015. [2]
- e. The population of Cambodia in 2015 was 15 677 000. Given that the increase in population from 2014 to 2015 was 1.68%, calculate the population of Cambodia in 2014. Give your answer correct to 3 significant figures. [3]

3. **S19/qp43/q1(c)**

There were 3.08×10^5 passengers that made this journey in 2018. This was a 12% decrease in the number of passengers that made this journey in 2017. Find the number of passengers that made this journey in 2017. [3]

4. **W20/qp43/q1 (0580)**

- a. The Earth has a surface area of approximately $510\,100\,000\text{ km}^2$
- Write this surface area in standard form. [1]
 - Water covers 70.8% of the Earth's surface. Work out the area of the Earth's surface covered by water. [2]
- b. The table shows the surface area of some countries and their estimated population in 2017.

Country	Surface area (km^2)	Estimated population in 2017
Brunei	5.77×10^3	433 100
China	9.60×10^6	1 388 000 000
France	6.41×10^5	67 000 000
Maldives	3.00×10^2	374 600

- Find the total surface area of Brunei and the Maldives. [1]
 - The ratio surface area of the Maldives : surface area of China can be written in the form $1 : n$. Find the value of n . [2]
 - Find the surface area of France as a percentage of the surface area of China. [2]
 - (iv) Find the population density of the Maldives. [Population density = population $\div$ surface area] [2]
- c. The population of the Earth in 2017 was estimated to be 7.53×10^9 . The population of the Earth in 2000 was estimated to be 6.02×10^9 .
- Work out the percentage increase in the Earth's estimated population from 2000 to 2017. [2]
 - Assume that the population of the Earth increased exponentially by $y\%$ each year for these 17 years. Find the value of y . [3]

Answer Key:

1. (a) Mercury, Mars, Venus, Earth
(b) 3000 or be 3×10^3
(c) $5.5(12) \times 10^{24}$.
(d) $1.09 \text{ to } 1.1(0) \times 10^{12}$.
2. (a) 138 404 000 or 1.38404×10^8 .
(b) Thailand
(c) $4.95[12] \times 10^7$
(d) 1.639 to 1.64
(e) 15 400 000
3. 3.5×10^5 .
4. (a)(i) 5.101×10^8 . (a)(ii) 361 150 800
(b)(i) 6070 (b)(ii) 32 000 (b)(iii) 6.68 or 6.677 (b)(iv) 1250 or 1248 to 1249
(c)(i) 25.1 or 25.08 (c)(ii) 1.33 or 1.325